Assignment

CO1 – Locomotion of Mobile Robots

1. **ARSHA I (TLY23CE016)**
   → *What are the key design challenges for a mobile robot?*
2. **GOKUL KRISHNA V (TLY23CE027)**
   → *Compare and contrast between wheeled robots and legged robots.*
3. **ABHISHEK S (TLY23CS005)**
   → *Explain a few applications of underwater robots.*
4. **ALWIN ALIYAS (TLY23CS019)**
   → *What are the key design challenges for a mobile robot?*

CO2 – Kinematic and Dynamic Modelling

5. **ANAMIKA P (TLY23CS023)**
   → *Explain the concept of degree of freedom and manoeuvrability.*
6. **BENEDICT JOSEPH (TLY23CS039)**
   → *Explain the kinematic model for a differential drive wheeled mobile robot.*
7. **FATHIMA SANVA T P (TLY23CS048)**
   → *Explain the dynamic modelling of a differential drive WMR using the Newton–Euler method.*
8. **HALEEMA ZAHA BINT NIHAD (TLY23CS055)**
   → *Explain the concept of degree of freedom and manoeuvrability.*

CO3 – Sensors for Mobile Robots

9. **HAMNA SHERIN C (TLY23CS056)**
   → *How are the sensors classified in mobile robotics?*
10. **MUHAMMED AJMAL P P (TLY23CS070)**
    → *Explain the working of ground-based beacon sensors.*
11. **MUHAMMED NOSHIN M (TLY23CS071)**
    → *Explain the working of vision-based sensors and their applications.*
12. **MUHAMMED SHARAS P K (TLY23CS072)**
    → *How are the sensors classified in mobile robotics?*

CO4 – Localization and Path Planning

13. **NAVDHA VASANTH (TLY23CS079)**
    → *What are the challenges faced in localization during the design of a robot?*
14. **N B ANSON ALEX (TLY23CS080)**
    → *Explain the Kalman method of map-based localization.*
15. **PRANAV PRADEEP (TLY23CS087)**
    → *Explain Dijkstra's algorithm for path planning.*
16. **RITHIKA P (TLY23CS092)**
    → *What are the challenges faced in localization during the design of a robot?*

CO5 – Obstacle Avoidance and Control

17. **SHAHEEN HASHIM (TLY23CS099)**
    → *Explain the Bug algorithm used for obstacle avoidance in mobile robots.*
18. **SNEHA R (TLY23CS109)**
    → *What is the Dynamic Window approach used for obstacle avoidance in mobile robots?*
19. **SOORAJ O O (TLY23CS110)**
    → *Using the kinematic model, explain any one method of control of a differential drive robot.*
20. **SREERAG P V (TLY23CS115)**
    → *Explain the Bug algorithm used for obstacle avoidance in mobile robots.*

CO6 – Applications of Mobile Robots

21. **THEJUS SREEJITH (TLY23CS122)**
    → *Explain the design considerations for the development of a differential drive robot moving to a specific point following a line.*
22. **VISWAS VINAYAN (TLY23CS128)**
    → *What are cooperative and collaborative robots?*
23. **VYSHNAV E G (TLY23CS129)**
    → *Write a brief note on mobile manipulators.*
24. **ROHAN A (TLY23EC103)**
    → *What are cooperative and collaborative robots?*
25. **RAYA SHUHAIB (TLY23IT057)**
    → *Explain the design considerations for the development of a differential drive robot moving to a specific point following a line.*